

Optimal Correction of Environmental Stochastic Externalities

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Abstract

How does pollution affect climate risk? The scientific literature views this question in two main ways. First, the "tipping point" view according to which pollution increases the probability of discrete catastrophic events. Second, the "distributional shift" view according to which pollution shifts the entire distribution of climate shocks and makes extreme weather events more frequent and more intense. In this paper, I study the implications of the second type of risk on the social cost of carbon (SCC), so far unexplored in the economic literature. I model what I call stochastic externalities in a framework with random climate shocks where the parameters of its distribution are functions of the pollution stock rather than fixed values. I solve the social planner's problem in a representative agent economy in a simple three-period model which I then generalize to an infinite-horizon recursive framework. The modified SCC sums a stochastic externality correction term to the standard Pigouvian term, which may be expressed in different ways but is essentially an expectation of future damages weighted by the distribution's sensitivity to climate variables. I provide a few illustrative examples: for Pareto shocks, the correction has a single tail-fattening channel which is an expectation of future damages weighted by an exponentially-growing term. This implies a stochastic externality correcting term which has the potential to grow faster without bound than the standard Pigouvian term, reinforcing Weitzman's concerns of a Dismal Theorem. In a simple numerical example, I show how the stochastic externality may significantly increase the SCC beyond the standard Pigouvian level. My framework allows for any parametric family of continuous distributions satisfying mild regularity conditions and nests the standard SCC as a special case, when shocks are "exogenous".

Keywords: *Optimal environmental taxation; stochastic externalities; carbon tax; endogenous climate risk; social cost of carbon; Pigouvian taxation*

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1 Introduction

How does pollution affect climate risk? The scientific literature views this question in two main ways. First, the "tipping point" view according to which pollution increases the probability of discrete catastrophic events. Second, the "distributional shift" view according to which pollution shifts the entire distribution of climate shocks and makes extreme weather events more frequent and more intense. These two views are different yet complementary: one is concerned with major regime shifts in the climate system that may happen in the future, while the other is concerned with the continuous increase of "everyday" climate risk, such as extreme events getting stronger and less predictable. Although a "tipping point" may be a civilizational threat, or indeed even an existential threat, the "distributional shift" perspective is nevertheless worth studying since the economy is already arguably significantly affected to this type of risk and will continue suffer from it in the future. Indeed, according to the NOAA, the number of billion-dollar weather disasters in the US has been increasing over the past decades, going from an average of 8.5 between 1980 and 2023 to 20.4 between 2017 and 2023 (NCEI, 2024). More recently, Bakkensen and Barrage (2026) empirically estimates the effect of climate change on future cyclone risk in the United States and the Caribbean using NOAA simulation data, finding that climate change will cause an almost threefold increase in welfare losses from cyclones in the United States.

In this paper, I develop a framework to analyze the implications of a "distributional shift" on the social cost of carbon (SCC). I model what I call stochastic externalities in a framework with random climate shocks where the parameters of its distribution are functions of the pollution stock rather than fixed values, as is common in DSGE economic modeling. First, to build intuition, I enhance a standard three-period neoclassical growth model where current emissions harm a fraction of future output and derive a SCC which sums a standard Pigouvian correction term and a stochastic externality correction term. There are many ways to express this second term but the conceptually simplest form may be the quantile form, where the expectation of future marginal damages is weighted by the change in the quantile function of the shock distribution caused by the increase in the pollution stock. In other words, marginal utility of damages is weighted by how much more or less likely a particular shock realization becomes after the pollution-induced distributional shift.

I then provide a few illustrative examples to show how this translates into more concrete economically-meaningful channels. In the normal distribution case, the stochastic externality correction decomposes into three channels: a mean-shift channel, a risk-aversion channel, and a damage-curvature channel. In the Pareto distribution case, the stochastic externality correction has a single tail-fattening channel which is an expectation of future damages weighted by an exponentially-growing term. This implies a stochastic externality correcting term which has the potential to grow faster without bound than the standard Pigouvian term, reinforcing Weitzman's concerns of a Dismal Theorem. In a simple numerical example, I show how the stochastic externality may significantly increase the SCC beyond the standard Pigouvian level. However, my framework also allows for any parametric family of continuous distributions satisfying mild regularity conditions and nests the standard SCC as a special case, when shocks are "exogenous".

Afterwards, to bridge theory and practice, I generalize this framework to a recursive infinite

horizon model. Although the standard recursive framework invokes Markov properties to argue for the validity of a recursive form (see for example Stokey et al., 1989), we can still involve Markov *Decision* Process arguments. Decision Markov processes are a generalization of Markov processes where the past state affects not only the current state, but also the probability distributions. Once the foundations are laid out, I show how this new "stochastic externality" channel affects the SCC in practice, modifying the DICE benchmark model to include a simple endogenous distributional shift effect. This example is illustrative and requires more realistic calibration, but nevertheless shows the practical applicability of this framework. Moreover, I use the normal distribution, which will by design imply a lower effect of stochastic externalities on the SCC, since it is thin-tailed and symmetrical. Further work will use more realistic calibrated thick-tailed distributions, drawing from both the scientific literature and the emerging microeconomic literature on the effects of weather extremes.

Before moving to the model, let us quickly summarize the current state of the literature to better position this work. This paper is not the first to model "endogenous" climate risk as there already exists a wealth of literature on the topic. For example, papers like Barro (2015), Brausmann and Bretschger (2024), Cai and Lontzek (2019), Müller-Fürstenberger and Schumacher (2015), van der Ploeg (2014), van der Ploeg and de Zeeuw (2018), and Zemel (2015) view climate change as affecting the probability of a particular catastrophic event occurring, for example by modeling shock occurrence as a Poisson process whose arrival rate increases with pollution. However, there is less guidance on how to address the increased prevalence and intensity of "everyday" extreme weather events, which is a matter not only of uncertain timing but also uncertain impacts. This paper fills this gap by modeling the entire distribution of climate shocks as shifting continuously with the pollution stock, thus capturing the gradual degradation of the climate risk profile rather than an increasing probability of a sudden regime shift. To my knowledge, this second mechanism that is now well explored in the Earth sciences has received much less attention in economics. However, some recent work is being done to estimate the effect of climate change on the distribution of weather shocks and their economic impacts, such as Bakkensen and Barrage (2026). Therefore, this paper further bridges the micro-macro gap by using the insights from the emerging empirical literature on the effects of climate change on the distribution of weather shocks as well as the implications on micro-level damages to inform the macro-level implications on optimal policy.

The remainder of the paper is organized as follows. Section 2 introduces the three-period model, progressing from the familiar standard externalities case to endogenous stochastic externalities and then ending with a normal distribution and a Pareto example. Section 3 generalizes the analysis to an infinite-horizon recursive model and Section 4 applies it to a modified DICE benchmark model. Finally, Section 5 concludes.

2 Three-period model

2.1 Standard externalities

Let's start with the simplest case, where there are only three periods and output Y contributes to a pollution stock S which reduces a fraction $D(S) \in [0, 1]$ of next period's total output. Suppose damages are increasing and convex in the pollution stock. The planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta u(c_1) + \beta^2 u(c_2) \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 - D(S_0))Y(k_0) \\ c_1 + k_2 - (1 - \delta)k_1 &= (1 - D(S_1))Y(k_1) \\ c_2 - (1 - \delta)k_2 &= (1 - D(S_2))Y(k_2) \\ S_1 &= (1 - \phi)S_0 + Y(k_0) \\ S_2 &= (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

This problem has the following modified Euler condition between periods 0 and 1.

$$u'(c_0) = \beta u'(c_1) \left(1 - \delta + \left[1 - D(S_1) - \underbrace{\beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)}_{\text{"Standard" externality}} \right] Y'(k_1) \right)$$

Since this is a standard problem, we do not solve the household's problem in detail. However, remember that the household's Euler condition does not contain the "standard" externality above and the planner would therefore like to set a carbon tax $\tau = \beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)$ on output to internalize the externality. This is the standard Pigouvian tax, equal to the discounted marginal damages of pollution. Since second-period output is entirely determined by first-period investment, only the Euler condition between periods 0 and 1 is affected.

We may also consider the case where damages are a function of the pollution stock as well as a random shock x following an "exogenous" distribution, $D(S, x)$. Suppose damages are increasing and convex in the pollution stock in climate shocks. In that case, we get the following Euler condition, implying an optimal carbon tax $\tau = \beta \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]$.

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(1 - \delta + \left[1 - D(S_1, x_1) - \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]}_{\text{"Standard" externality}} \right] Y'(k_1) \right) \right]$$

2.2 Stochastic externalities

Now consider the same model where future damages are stochastic, but the parameters of their distribution shifts with the pollution stock. We make the following hypotheses with regards to the stochastic process: (i) the shocks follow a statistical model of the parametric family which (ii) has fixed support, (iii) a pdf and cdf continuous and differentiable in parameter θ , (iv) a parameter θ continuous and differentiable in c_0 , and (v) a strictly increasing cdf. We represent the model's dependence on current consumption by treating the parameter $\theta(c_0)$ as a function of past consumption. The social planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta(S_1)) dx_1 \\ & + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta(S_1)) f(x_2; \theta(S_2)) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned} \tag{1}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 - D(S_0))Y(k_0) \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= (1 - D(S_1, x_1))Y(k_1) \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= (1 - D(S_2, x_2))Y(k_2(x_1)) \quad \forall x_1, x_2 \\ S_1 &= (1 - \phi)S_0 + Y(k_0) \\ S_2 &= (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

We write the expectations in integral form to make the dependence of the distribution on the pollution stock explicit which emerges in the PDF $f(x; \theta(S))$. Meanwhile, the consumer solves the following problem, where we denote by θ_1 and θ_2 the parameters of the distributions for x_1 and x_2 respectively.

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta_1) dx_1 + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 + r_0)k_0 \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= r_1(x_1)k_1 \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= r_2(x_2)k_2(x_1) \quad \forall x_1, x_2 \end{aligned}$$

The consumer takes the PDF as exogenous, hence us not treating θ as functions of pollution. This leads to the following modified expressions for the optimal corrective tax.

Proposition 2.2.1 (Three-period carbon tax with endogenous stochastic externalities).
The planner can decentralize the solution to problem (1) with the following carbon tax:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

Proof. See Appendix A for full proof. □

Looking at the formulas, we notice that our set up nests the standard externality. Indeed, we get the deterministic Pigouvian tax when the shocks follow a degenerate distribution and the Pigouvian tax in expectations when the shocks follow an exogenous distribution. With stochastic externalities, we get equation (4) with an additional term which implies a higher corrective tax when it is negative.

The first equation in Proposition 2.2.1 has a positive second term when utility and the score move in the same direction on average. We know that utility is monotone decreasing in shocks since higher shocks mean lower output and therefore lower consumption. Meanwhile, the score can be positive or negative depending on whether the probability of a certain shock realization increases or decreases after the probability shift.

The second equation is equivalent, but has a different interpretation. The first term is the stochastic discount factor, the second term is the marginal damages of shocks on output, and the third term is the change in the quantile function caused by the probability shift. Intuitively, if we for example think of the 3rd quartile, this says whether the position of this quartile goes up (75% of shocks are now higher) or down (75% of shocks are now lower).

To build some intuition on how the extra term works, we present two illustrative examples. Nevertheless, the framework we present is generalizable to any distribution as long as the distribution satisfies the assumptions mentioned at the start of the section.

Normal shocks. Let $x \sim N(\mu(S), \sigma(S))$ such that $\mu'(S) > 0$ (higher pollution increases average shocks) and $\sigma'(S) > 0$ (higher pollution increases shock variability).

Proposition 2.2.2 (Normal distribution decomposition).

Under normal shocks, the stochastic externality correction term decomposes into 3 channels:

$$\begin{aligned} -\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad - \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

Proof. See Appendix B for full proof. □

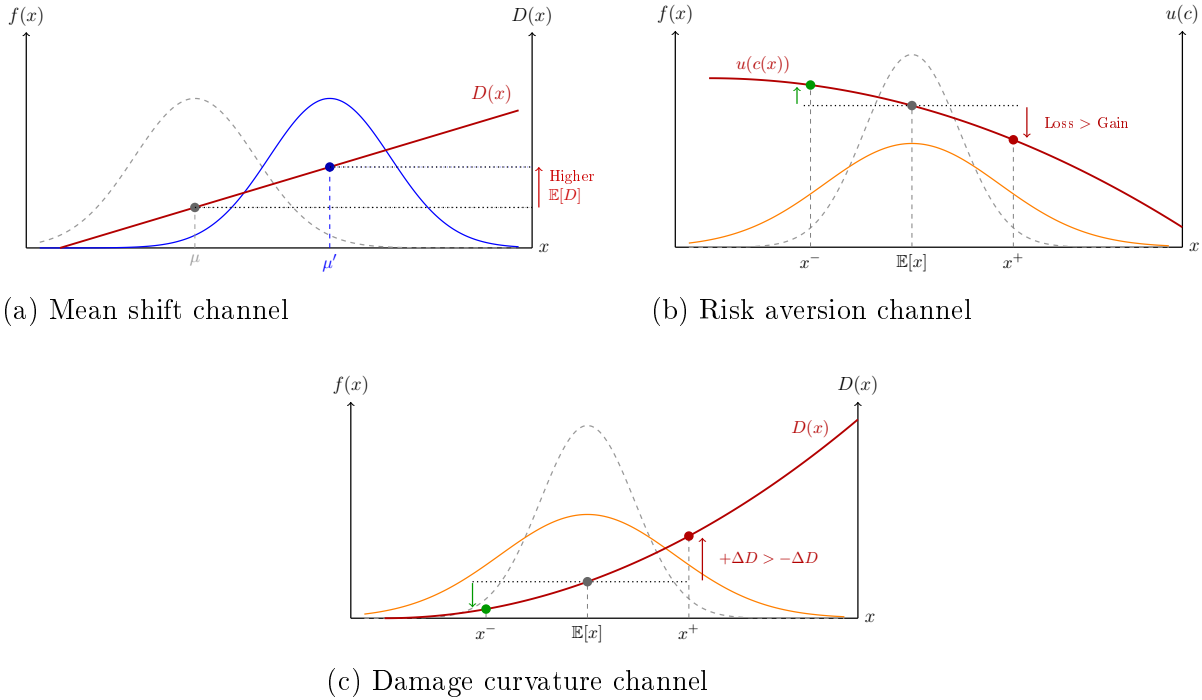


Figure 1: Stochastic externality channels for normally distributed shocks.

Figure 1 gives an intuition for the three channels. The mean shift channel in panel (a) leads to a higher carbon tax if damages are increasing since climate shocks are now higher on average. The risk aversion channel in panel (b) leads to a higher carbon tax if utility is concave because, although a variance increase makes both lower and higher shocks more likely, this causes a higher utility decline in the right-tails than utility gained in the left-tails.

If utility is linear such that the agent is risk-neutral, the risk aversion channel vanishes because both extremes become equally likely, yet the agent is indifferent between them. The damage curvature channel in panel (c) depends on the curvature of the damage function with respect to climate shocks. If damages are convex, meaning damages grow faster as shocks grow larger, this channel raises the tax because expanding the tails increases damages on the right tail more than it decreases damages on the left tail.

Pareto shocks. Let $x \sim \text{Pareto}(\alpha(S), x_m)$ with $\alpha'(S) < 0$ (higher pollution reduces the shape parameter, making the tail heavier) and fixed support $x_m > 0$. Moreover, suppose $\alpha > 1$ such that a finite first moment exists.

Proposition 2.2.3 (Pareto distribution decomposition).

Under Pareto shocks, the stochastic externality correction term can be rewritten as follows:

$$-\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{D(\cdot)}{\partial x_2} Y(k_2) x_2 \ln \left(\frac{x_2}{x_m} \right) \right]$$

Proof. The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ and CDF is $F(x; \alpha) = 1 - \left(\frac{x_m}{x}\right)^\alpha$ for $x \geq x_m$. We directly apply the result in 2.2.1 on the CDF form of the stochastic externality term. $\square$

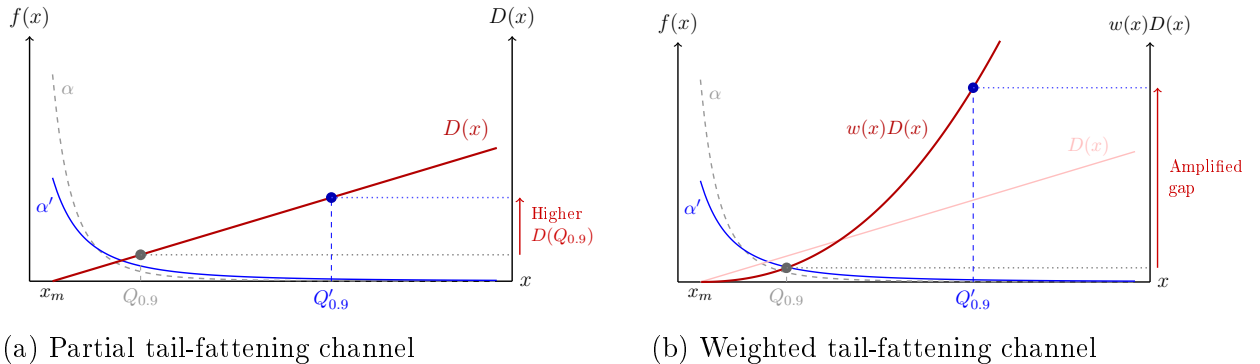


Figure 2: Pareto tail-fattening channel.

Endogenous Pareto shocks involve a single tail-fattening channel, as illustrated in figure 2. When the shape parameter α decreases and the right tail becomes heavier, each quantile of shocks shifts right as shown in panel (a). However, as shown in panel (b), right-tail damages are also weighted by a quantile shift term $w(x) \equiv x \ln(x/x_m)$ which grows exponentially, putting an increasing weight on rare catastrophic events. Depending on functional forms, the above expectation might diverge and imply an unbounded stochastic externality term, echoing the "Dismal Theorem" in Weitzman (2009). Indeed, the expectation is more "fragile" than the standard externality term. Since this paper is not focused on the Dismal Theorem, we will limit ourselves to a quick detour. Let us focus on $x \ln(x/x_m)$ and ignore the other terms in the expectation. If $\alpha > 1$, we have $\mathbb{E}[x] = \frac{\alpha x_m}{\alpha - 1}$. Now let us turn to the expectation of $x \ln(x/x_m)$ and note that $y \equiv \ln(x/x_m)$ follows an exponential distribution with rate α .

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = x_m \mathbb{E}[ye^y]$$

$\mathbb{E}[ye^y]$ is the derivative of the exponential distribution's moment generating function with respect to the moment parameter. Indeed,

$$M_y = \mathbb{E}[e^{sy}] \implies \frac{\partial M_y}{\partial s} = \mathbb{E}[ye^{sy}]$$

Since we can also write $M_y = \alpha/(\alpha - s)$,

$$M_y = \frac{\alpha}{\alpha - s} \implies \mathbb{E}[ye^y] = \frac{\partial M_y}{\partial s} \Big|_{s=1} = \frac{\alpha}{(\alpha - 1)^2}$$

Hence,

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = \frac{\alpha x_m}{(\alpha - 1)^2} = \frac{\mathbb{E}[x]}{\alpha - 1}$$

As $\alpha \rightarrow 1$ and the tails become heavier, $\mathbb{E}[x \ln(x/x_m)]$ diverges faster than $\mathbb{E}[x]$ such that the stochastic externality term may become unbounded before the standard externality term. Hence, even if we find that the Dismal Theorem does not hold with standard externalities¹, it may still hold with stochastic externalities. We leave further analysis of the implications of stochastic externalities on the Dismal Theorem for future work. However, the main takeaway is that the assumption on the distribution of shocks affects not only the channels by which stochastic externalities operate but also their importance relative to standard externalities, having the potential to dominate them.

In the next section, we generalize the simple three-period framework to an arbitrary number of periods. Before we move on, we want to stress that our general results are not a consequence of the specific way we have modeled the externality. We chose the standard neoclassical model with multiplicative damages on output to better build intuition, but we could have very well had modeled externalities as emerging from polluting consumption instead of polluting output. Though the specific form of the externality would be different, a stochastic externality would still arise with the same channels at play².

¹See for example Nordhaus (2009) or Anthoff and Tol (2022) which present the Dismal Theorem as a real concern but show its applicability depends on the hypotheses being made and the modeling context.

²For example, polluting consumption would affect the Euler condition through marginal utility rather than marginal product of output. However, we would still add a standard externality term and a stochastic externality term, with its channels depending on the assumption on the distribution of shocks.

3 Multiple period model

Now suppose an infinite horizon problem and suppose a pollution stock accumulates, with the past stock depreciating linearly at rate ϕ . In integral form, the sequential problem can be written in the following way, where the pdf of shocks depends on the pollution stock. Let $x^t \equiv (x_0, x_1, \dots, x_t)$ be the history of shocks up to t and $\beta \in (0, 1)$ be the discount factor.

$$\max_{\{c_t(x^t), k_{t+1}(x^t), S_{t+1}(x^t)\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \int \cdots \int u(c_t(x^t)) f(x_0; \theta(S_0)) \cdots f(x_t; \theta(S_t)) dx_0 \cdots dx_t \right\} \quad (2)$$

s.t.

$$\begin{aligned} c_t(x^t) + k_{t+1}(x^t) &= (1 - D(S_t, x_t))Y(k_t(x^{t-1})) + (1 - \delta)k_t(x^{t-1}) & \forall t \\ S_{t+1}(x^t) &= (1 - \phi)S_t(x^{t-1}) + Y(k_t(x^{t-1})) & \forall t \\ x_{t+1} &\sim f(x_{t+1}; \theta(S_{t+1}(x^t))) & \forall t \end{aligned}$$

As written, this sequential problem is not very tractable. Hence, it can be useful to rewrite it recursively to make it more akin to the two-period problem. Although standard recursive methods as often used in economics assume exogenous shocks following a Markov process (see for example Stokey, Lucas, and Prescott (1989)), recursivity can still be established here. Indeed, our shocks essentially follow a *Markov decision process* (MDP) where the transition probability depends on the current state and the current action. With minor adjustments, the main results of SLP still hold³. First, let us make the following standard assumptions.

Assumption 3.0.1.

Let Γ be the correspondence for the constraints of this problem mapping states $s \in \mathbb{S}$ (i.e. capital and pollution stock) to actions $a \in \mathbb{A}$ (i.e. output). Suppose Γ has a measurable selection, meaning there is a measurable function $h : \mathbb{S} \rightarrow \mathbb{A}$ such that $h(s) \in \Gamma(s) \quad \forall s \in \mathbb{S}$.

Assumption 3.0.2.

Let utility $u : \mathbb{S} \times \mathbb{A} \rightarrow \mathbb{R}$ and the stochastic kernel $P : (\mathbb{S} \times \mathbb{A}) \times \mathcal{S} \rightarrow [0, 1]$, where $\mathcal{S}$ is the σ -algebra over the state space $\mathcal{S}$. Let a plan $\pi \in \Pi$ be a sequence of actions $(\pi_0, \pi_1, \dots)$ such that $\pi_0 \in \mathbb{A}$ and $\pi_t : \mathcal{S}^t \rightarrow \mathbb{A}$ are measurable for all t . For each $s_0 \in \mathbb{S}$ and each $\pi \in \Pi(s_0)$, $u(s_t, \pi_t(s^t))$ is $\mu^t(s_0, ds^t; \pi)$ -integrable and the following limit exists.

$$u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi)$$

³See Miao (2013) for a simple introduction to MDP dynamic programming or, for more formal treatment, Bertsekas (1995), Hernández-Lerma and Lasserre (1996), etc.

Assumption 3.0.3.

Let there be a collection of non-negative measurable functions $L_t = \mathbb{S} \rightarrow \mathbb{R}_+ \forall t$ such that

$$u(s_0, \pi_0) \leq L_0(s_0) \text{ , } u(s_t, \pi_t(s^t)) \leq L_t(s_0) \text{ , } \sum_{t=0}^{\infty} \beta^t L_t(s_0) < \infty \quad \forall \pi \in \Pi(s_0), s_0 \in \mathbb{S}$$

Assumption 3.0.1 ensures a feasible policy exists for any state, Assumption 3.0.2 ensures that the discounted sum of utilities is well-defined, and Assumption 3.0.3 is a technical condition allowing for the use of the dominated convergence theorem in the following lemma.

Lemma 3.0.1 (Recursive formulation).

We can rewrite the sequential problem (2) as the following recursive problem.

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & s.t. \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned} \tag{3}$$

Proof. We follow Miao (2013) for this proof, but other textbooks may be used as reference. It suffices to show that our setting is a standard MDP dynamic programming problem. Here, the state is the triple (k, S, x) and the action is (k', S') . Note that we can rewrite the constraints such that c is an implicit function of (k', S') . By Assumptions 3.0.1 and 3.0.2, we can define the following discounted expected utility for any given state s_0 and policy π .

$$J(s_0, \pi) = u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \left\{ \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi) \right\}$$

Moreover, for any state, we can define the following value function.

$$V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$$

With our particular notation, $s_0 = (k_0, S_0, x_0)$, $s_t = (k_t, S_t, x_t) \forall t$, and $s = (k, S, x)$. Here, $\pi \in \Pi(s)$ means that the policy π is feasible given the initial state s , where feasibility is established by the constraint set.

Suppose that $V : \mathbb{S} \rightarrow \mathbb{R}$ is measurable and, given any $s \in \mathbb{S}$ and any $\epsilon > 0$,

$$\exists \pi^s \in \Pi(s) : J(s, \pi) + \epsilon \geq V(s)$$

such that π_0^s and $\pi_t^s(s_1^t)$ are measurable for all t . By Theorem 7.1.1 of Miao (2013), Assumptions 3.0.1-3.0.3 together with the above assumption means V satisfies:

$$V(s) = \sup_{a \in \Gamma(s)} \left\{ u(s, a) + \beta \int V(s') P(s, a; ds') \right\}$$

In our notation,

$$V(k, S, x) = \max_{(k', S') \in \Gamma(k, S, x)} \left\{ u((k, S, x)) + \beta \int V(k', S', x') f(x'; S') dx' \right\}$$

Alternatively, noting that c is an implicit function of (k, S, x) ,

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & \text{s.t.} \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned}$$

By Theorem 7.1.2 of Miao (2013), this equation is also equivalent to $V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$, which means that the recursive problem is equivalent to the sequential problem. $\square$

Hence, although the shocks appear i.i.d., the dependence on last period's emissions creates persistence in the stochastic process. Since we assumed that consumption only affects the distribution of shocks through the pollution stock, the state variable S is sufficient to capture the effect of past emissions on future shocks. This gives the stochastic process a Markov structure, which allows us to write the problem in a recursive form. Using this result, we can derive the Euler condition for the multi-period model, generalizing the results from the two-period model to an arbitrary horizon.

Proposition 3.0.1 (Carbon tax with endogenous stochastic externalities).

The planner can decentralize the solution to problem (3) with the following carbon tax:

$$\tau_t = \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} - \underbrace{\beta \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\begin{aligned} \tau_t = & \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} \\ & + \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right]}_{\text{"Stochastic" externality}} \end{aligned}$$

Proof. See Appendix C for full proof. □

The above formula generalizes the three-period case to an arbitrary horizon. Because the polluting stock depreciates at rate ϕ , the effect of current emissions on future shocks is discounted by a factor $(1 - \phi)^j$ after j periods. That is, only $(1 - \phi)^j$ of emissions at time t will remain and harm the environment at time $t + j$. The first term is the standard Pigouvian term equal to the discounted expected marginal damages of pollution, while the second term is the "risk endogeneity" term capturing the effect of pollution on the distribution of shocks.

4 DICE-SE model

To illustrate the potential importance of stochastic externalities, I enhance the benchmark DICE model first developed in Nordhaus (2009) and later iterations, which I henceforth call DICE-SE, where "SE" stands for "stochastic externalities". Although other integrated assessment models (IAMs) have been developed, many with more complex climate and economic dynamics yielding higher SSCs, the DICE model remains a useful benchmark for its simplicity and transparency. Moreover, the main purpose of this exercise will be to compare the SSC with and without stochastic externalities, so the specific features of the DICE model are not crucial for our analysis.

To ensure that the results are not driven by the specification or calibration of a particular DICE model, I build upon the DICE model as written in Cai and Lontzek (2019)⁴. While their original code is written in GAMS, a proprietary programming language, I first rewrite it in Julia. Then, I add the stochastic externality channel.

The base DICE model is identical to the DICE-2007 model, as in Cai and Lontzek (2019). Gross output follows a Cobb-Douglas production function:

$$Y_t^{\text{gross}} = A_t \cdot L_t^{1-\gamma} \cdot K_t^\gamma$$

where A_t is total factor productivity (TFP) and L_t is population. Total emissions are:

$$E_t = \sigma_t(1 - \mu_t)Y_t^{\text{gross}} + E_t^{\text{land}}$$

and enter the carbon cycle in GtC directly. The three-box carbon cycle, two-layer climate model, and capital accumulation follow the standard equations in the base model. The climate model uses current-period M^{AT} for forcing:

$$T_{t+1}^{\text{AT}} = T_t^{\text{AT}} + c_1 \left(F_{2 \times \text{CO}_2} \log_2 \frac{M_t^{\text{AT}}}{M_{\text{eq}}} + F_t^{\text{oth}} - \lambda T_t^{\text{AT}} - c_3(T_t^{\text{AT}} - T_t^{\text{OC}}) \right)$$

The model uses Weitzman-style damages:

$$\Omega_t^{\text{det}} = a_1 T_t^{\text{AT}} + a_2 (T_t^{\text{AT}})^2$$

and net output is

$$Y_t^{\text{net}} = \frac{Y_t^{\text{gross}}}{1 + \Omega_t} - \text{ABATECOST}_t$$

For illustrative purposes, I will add a stochastic damage component to the originally deterministic damage function as follows. At each period, a climate shock x_t is drawn from a normal distribution whose parameters depend on the atmospheric CO_2 concentration:

⁴Although the authors solve an enhanced DSICE model, their replication package also features code for the standard DICE model, as a comparison. We use this code and not the DSICE code so that there is a separation with Cai and Lontzek's additions.

$$x_t \sim \mathcal{N}(\mu_x(s_t), \sigma_x(s_t)^2)$$

where $s_t = (M_t^{\text{AT}} - M_{\text{eq}})/M_{\text{eq}}$ is normalized excess CO₂, and

$$\begin{aligned}\mu_x(s) &= \mu_1 \cdot s, \\ \sigma_x(s) &= \sigma_0 \cdot e^{\sigma_1 \cdot s}.\end{aligned}$$

The stochastic damage fraction is:

$$\Omega_t = \Omega_t^{\text{det}} \cdot (1 + x_t)$$

This specification implies that as atmospheric CO₂ rises above preindustrial levels ($s > 0$): (i) the mean damage shock shifts upward ($\mu_x > 0$), increasing expected damages, and (ii) the variance of damage shocks grows (σ_x increases), amplifying climate risk.

In words, damages will fluctuate around what would normally be the deterministic damage function. Using a normal distribution is simple, yet perhaps unrealistic given that extreme events are often assumed to have heavy-tailed or fat-tailed distributions such as Pareto, lognormal, Generalized Extreme Value, etc. Nevertheless, the normal distribution provides a conservative benchmark and indeed perhaps a lower bound on what the impact of stochastic externalities on the SCC could be. For the time being, I will not attempt to determine the appropriate functional form for the distribution of shocks, nor to calibrate the parameters of this distribution. Instead, I will simply choose some illustrative values for these parameters and show how the SCC changes with and without stochastic externalities. In future work, I will attempt to determine the appropriate functional form and calibrate the parameters using empirical data on climate damages.

5 Results

5.1 Calibration and scenarios

I compare three scenarios. The *deterministic* scenario sets $\sigma_0 = 0$, eliminating all stochastic shocks and recovering the standard deterministic DICE model. The *exogenous stochastic* scenario draws damage shocks from a fixed normal distribution independent of the CO₂ stock ($\mu_1 = \sigma_1 = 0$, $\sigma_0 = 0.10$), corresponding to standard DICE models with random damages. The *endogenous stochastic* scenario activates the full stochastic externality channel, with both the mean and variance of damage shocks depending on atmospheric CO₂. Any difference in optimal policy between the exogenous and endogenous scenarios isolates the stochastic externality: the additional incentive for carbon abatement that arises because pollution shifts the damage distribution.

The shock parameters are set to illustrative values: $\mu_1 = 0.05$, $\sigma_0 = 0.10$, and $\sigma_1 = 0.10$. Given the initial normalized excess CO₂ of $s_0 = (M_0^{\text{AT}} - M_{\text{eq}})/M_{\text{eq}} = (808.9 - 596.4)/596.4 \approx 0.356$, the initial mean damage shock is $\mu_x(s_0) \approx 1.8\%$ and the baseline standard deviation is $\sigma_x(s_0) \approx 10.4\%$. All remaining parameters follow the DICE-CJL calibration of Cai and Lontzek (2019). The model is solved by backward induction on a Smolyak sparse grid over the six-dimensional state space $(K, M^{\text{AT}}, M^{\text{UP}}, M^{\text{LO}}, T^{\text{AT}}, T^{\text{OC}})$, using Chebyshev polynomial approximations for the value function and Gauss-Hermite quadrature to integrate over the damage shock distribution.

5.2 The stochastic externality premium

Table 1 and Figure 3 present the central result: the optimal carbon price at the initial period (year 2000) for each scenario. In the DICE framework, the marginal abatement cost at the optimum equals the social cost of carbon, so the initial-period carbon price serves as a direct measure of the SCC under each regime.

Table 1: Carbon Price at Initial Period (SCC Proxy): Scenario Comparison

Scenario	Carbon price (\$/tCO ₂)	Premium (\$/tCO ₂)	Premium (%)
Deterministic	39.20	—	—
Exogenous stochastic	39.19	-0.01	-0.0%
Endogenous stochastic	44.01	+4.81	+12.3%
<i>Stochastic externality (Endo – Exog)</i>		+4.82	+12.3%

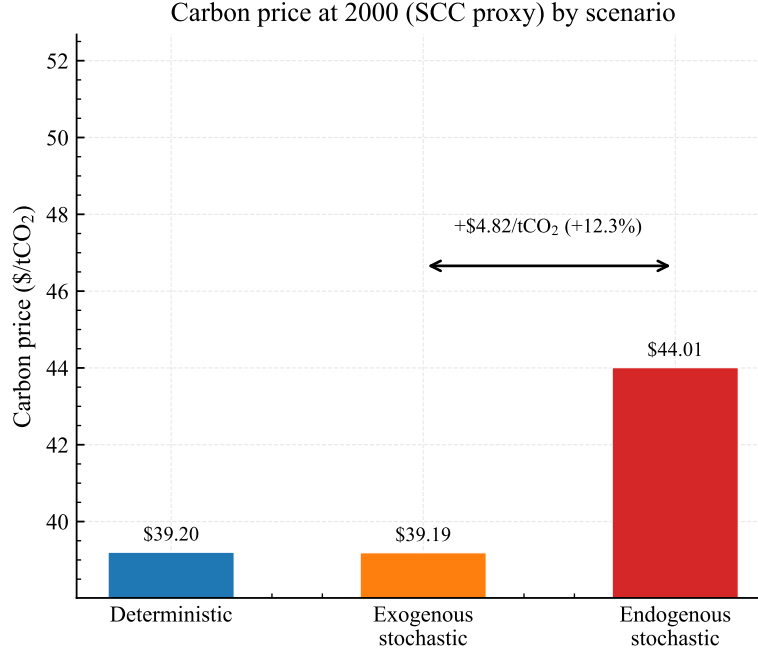


Figure 3: Initial-period optimal carbon price (year 2000) across scenarios. The double-headed arrow highlights the stochastic externality premium: the difference between the endogenous and exogenous stochastic scenarios.

The most striking result is the near-perfect equivalence of the deterministic and exogenous stochastic scenarios. Although the exogenous scenario introduces randomness into the damage function, the optimal carbon price is almost unchanged relative to the deterministic benchmark (\$39.20 vs. \$39.19 per tCO₂). This finding reflects a fundamental property of the standard model: when the damage shock distribution is independent of the CO₂ stock, there is no distributional externality to internalize. Randomness alone has little effect on the first-best carbon price.

The endogenous scenario tells a sharply different story. When pollution shifts both the mean and variance of damage shocks, the optimal carbon price rises to \$44.01 per tCO₂, a 12.3% premium over the exogenous benchmark. This is the stochastic externality: the additional carbon tax required to correct for the effect of current emissions on the future riskiness of the climate. Under CRRA utility with elasticity $\eta = 2$, the planner internalizes both higher expected damages (the mean-shift channel) and greater damage volatility (the variance channel), and the optimal policy reflects both motives simultaneously. Note that the precise value of the premium is illustrative and will depend on the functional form of the shock distribution as well as its parameters.

The stochastic externality premium is not a static initial-period phenomenon: it grows in absolute terms throughout the simulation horizon. Figure 4 shows the optimal carbon price paths for all three scenarios over 2000-2100.

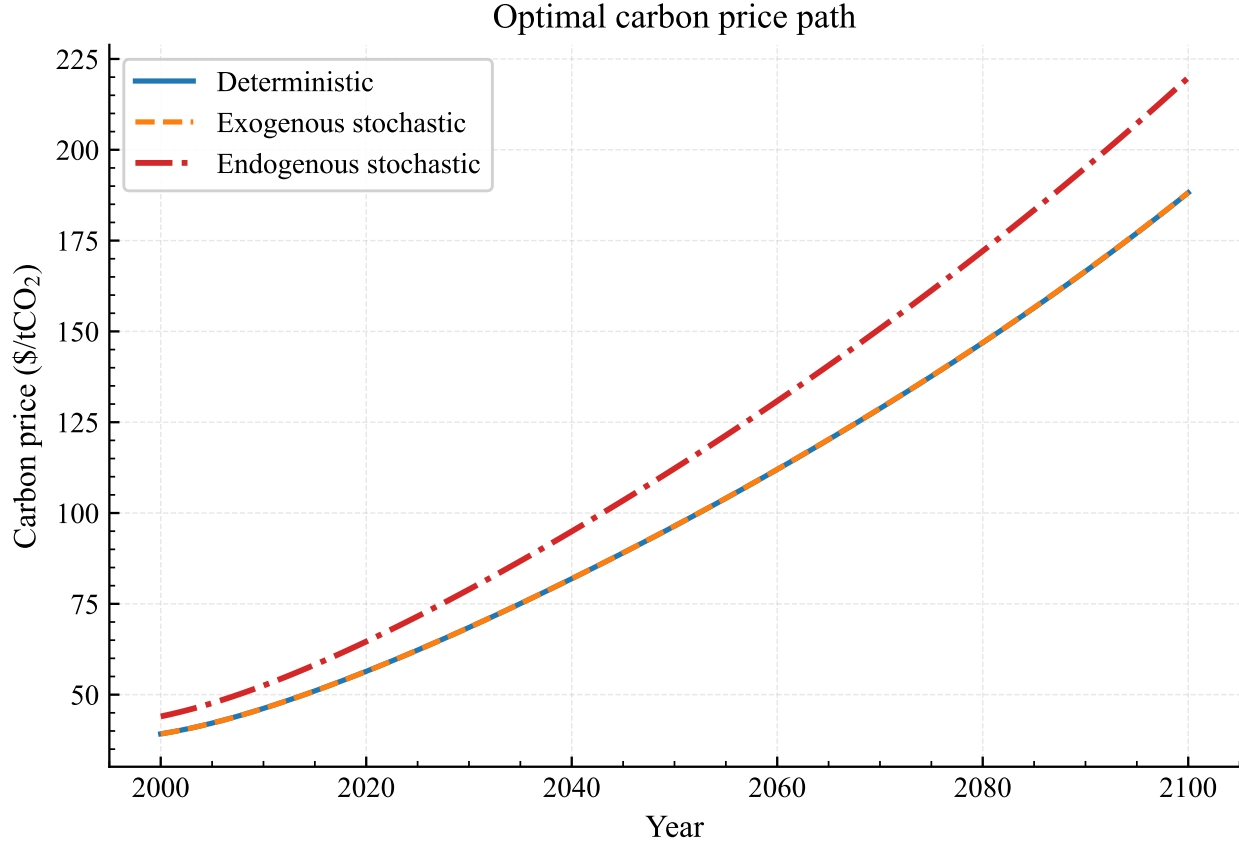


Figure 4: Optimal carbon price (\$/tCO₂) under three scenarios, 2000–2100. The endogenous stochastic scenario prescribes a persistently higher carbon price than both benchmarks, with the gap widening over time.

The deterministic and exogenous paths remain virtually indistinguishable throughout the century, confirming that exogenous stochasticity has negligible policy consequences. The endogenous path lies persistently above both, with the absolute premium widening from \$4.82 per tCO₂ in 2000 to \$9.29 in 2025, \$15.82 in 2050, \$23.55 in 2075, and \$31.61 in 2100. In percentage terms, the premium rises from 12.3% at the initial period to approximately 14–17% over the remainder of the century. This escalation is driven by CO₂ accumulation: as the atmospheric carbon stock grows, s_t increases, amplifying both $\mu_x(s_t)$ and $\sigma_x(s_t)$ and hence the marginal cost of emitting an additional tonne of carbon today.

Figure 5 isolates the stochastic externality premium directly. Since the exogenous and deterministic paths are nearly identical, the two premium series overlap almost perfectly, confirming that the entire wedge between the endogenous and deterministic cases is attributable to the stochastic externality channel.

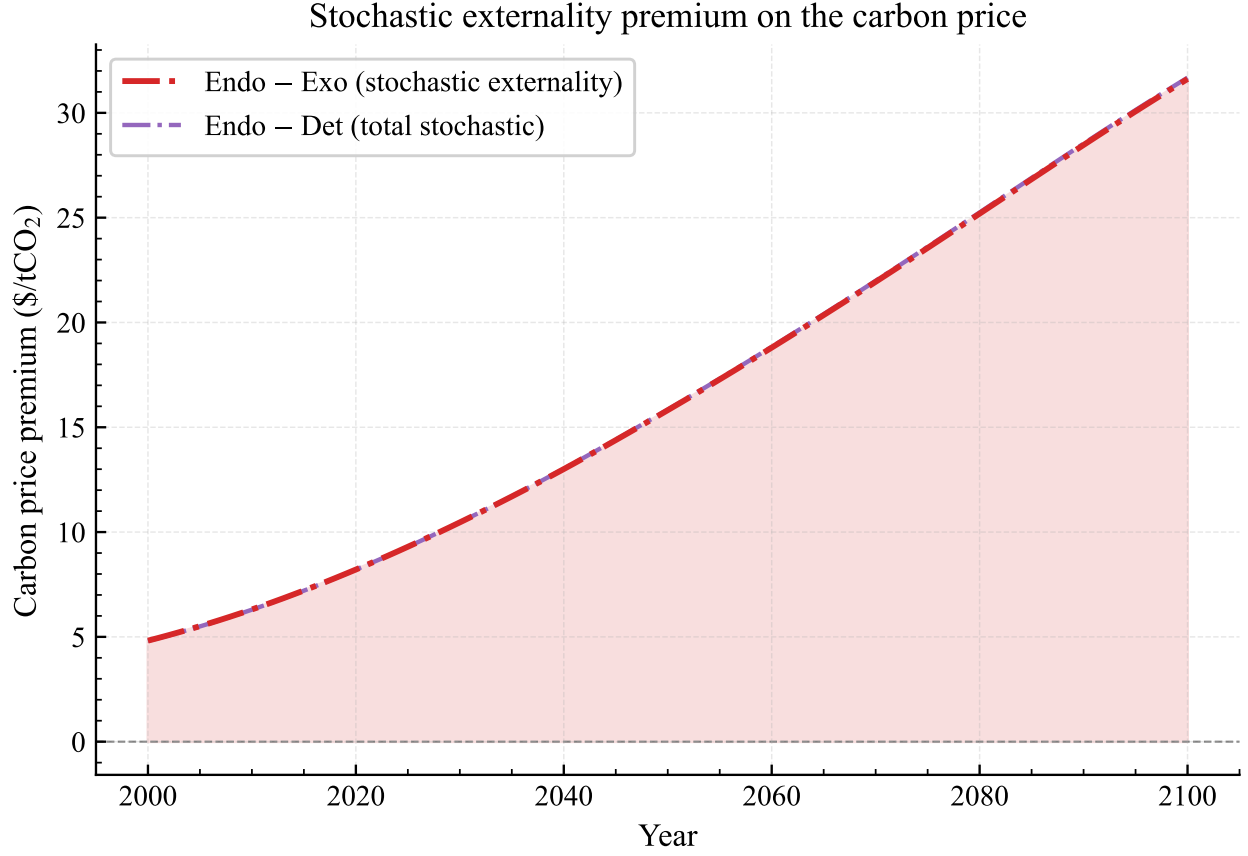


Figure 5: Carbon price premium of the endogenous stochastic scenario over the exogenous (stochastic externality) and deterministic benchmarks, 2000–2100. The shaded region highlights the stochastic externality premium (Endo – Exo).

The higher carbon price under the endogenous scenario translates into more aggressive mitigation. Figure 6 shows the optimal mitigation rate μ_t for each scenario. The endogenous scenario achieves substantially higher abatement throughout: by 2025, 21.9% of baseline emissions are abated versus 20.3% in the benchmarks, a gap of 1.6 percentage points. This gap widens to 2.3 points by 2050, 3.0 points by 2075, and 3.7 points by 2100. By 2100, the endogenous scenario abates 44.6% of baseline emissions versus 40.9% in the benchmarks.

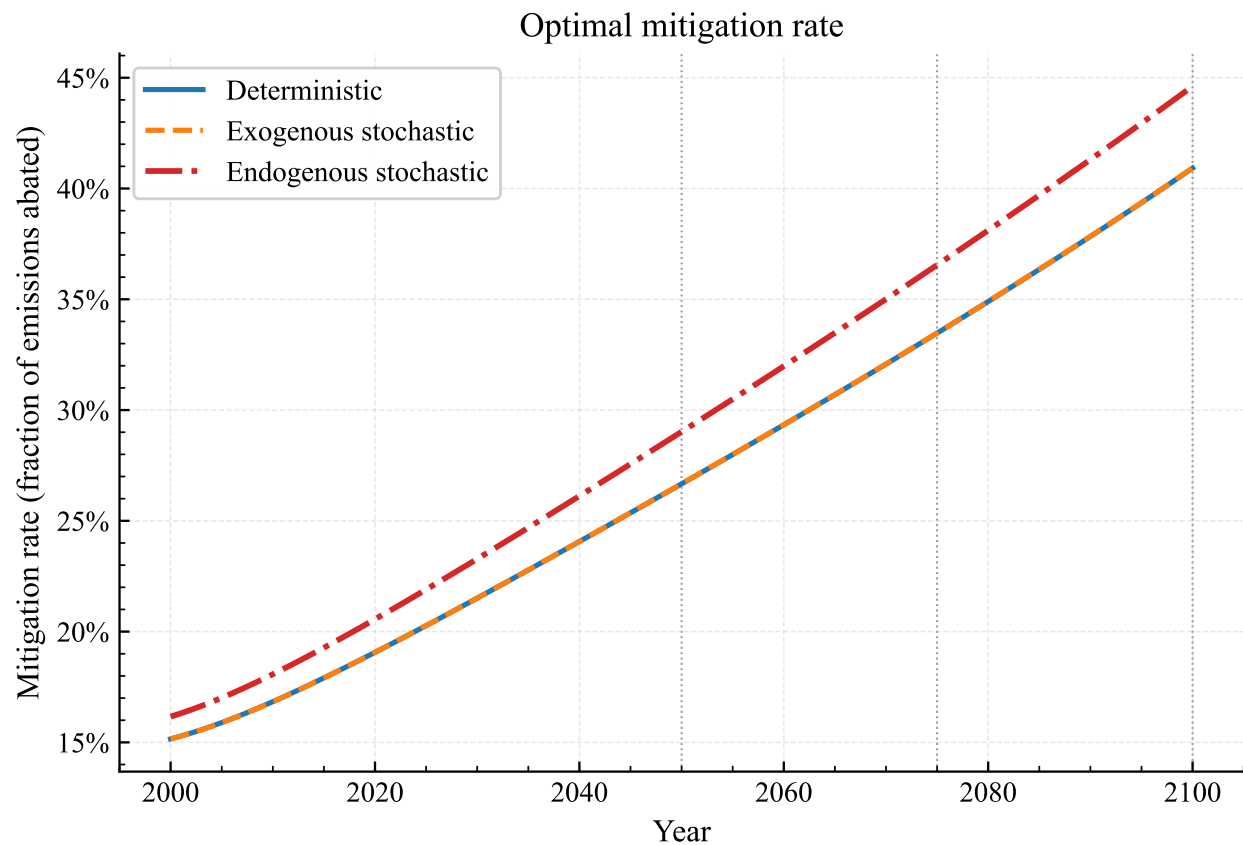


Figure 6: Optimal mitigation rate (fraction of baseline emissions abated) under three scenarios, 2000–2100.

Table 2 summarizes key outcome variables at benchmark years for each scenario.

Table 2: Key variables at selected years across scenarios

Year	Scenario	TATM (°C)	CPRICE (\$/tCO ₂)	MIU	CPC (k\$/person)
2025	Deterministic	1.215	62.27	20.3%	8.35
	Exogenous stochastic	1.215	62.27	20.3%	8.35
	Endogenous stochastic	1.212	71.56	21.9%	8.35
2050	Deterministic	1.704	96.41	26.7%	11.46
	Exogenous stochastic	1.704	96.43	26.7%	11.46
	Endogenous stochastic	1.691	112.25	29.0%	11.46
2075	Deterministic	2.153	137.65	33.5%	15.51
	Exogenous stochastic	2.153	137.71	33.5%	15.51
	Endogenous stochastic	2.126	161.26	36.5%	15.51
2100	Deterministic	2.557	188.17	40.9%	20.78
	Exogenous stochastic	2.557	188.24	40.9%	20.78
	Endogenous stochastic	2.513	219.85	44.6%	20.78

The more aggressive mitigation under the endogenous scenario produces measurably lower atmospheric warming and CO₂ concentrations. Figure 7 shows the temperature and CO₂ trajectories over 2000–2150.

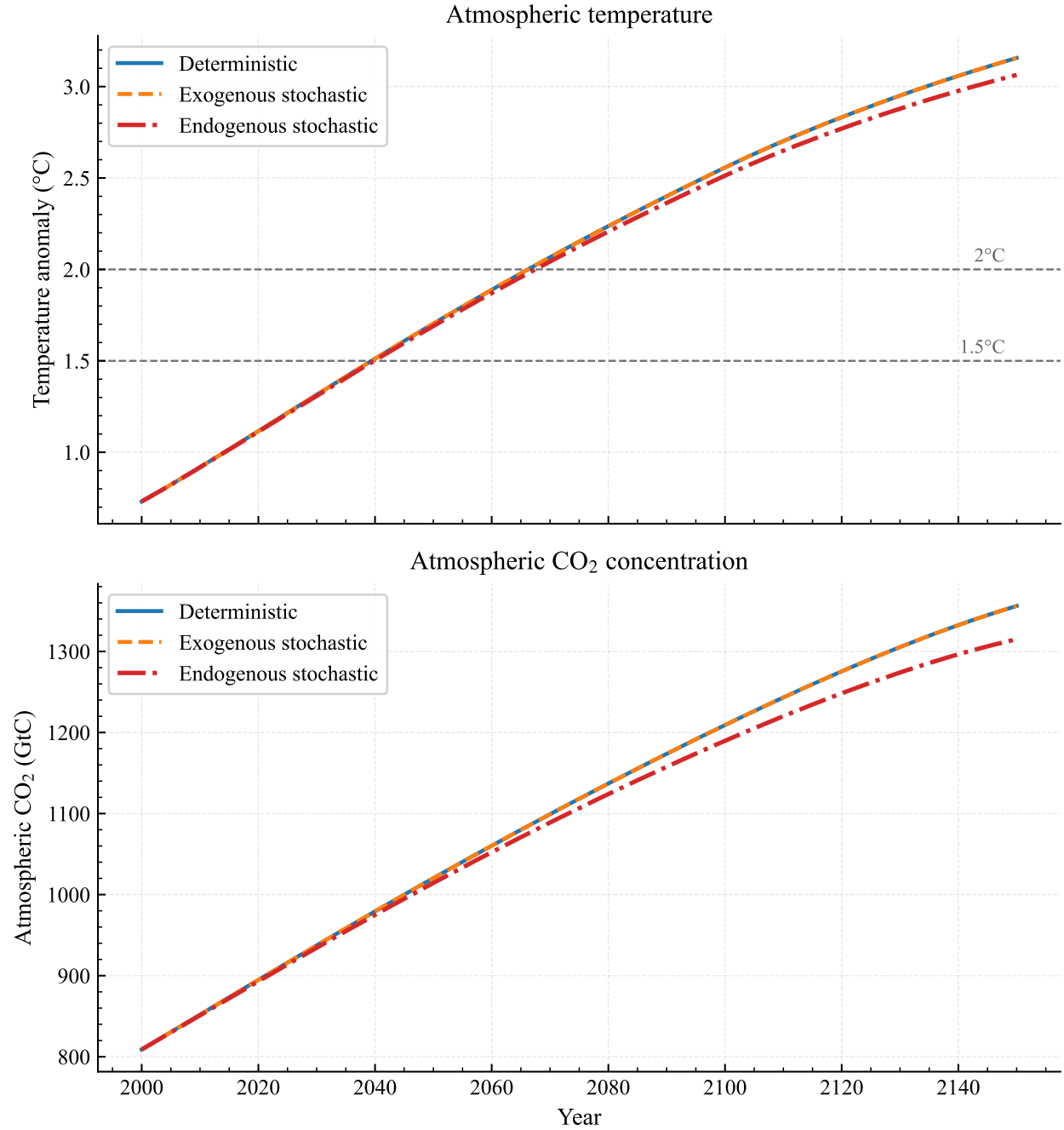


Figure 7: Atmospheric temperature anomaly (top) and atmospheric CO₂ (bottom) under three scenarios, 2000–2150. Horizontal dashed lines in the top panel mark the 1.5°C and 2°C Paris Agreement thresholds.

The temperature differential between the endogenous and benchmark scenarios starts small, roughly 0.003°C in 2025, but compounds to 0.013°C by 2050 and 0.044°C by 2100, with further divergence thereafter. Under all three scenarios, atmospheric warming crosses the 1.5°C threshold shortly after 2025 and approaches the 2°C threshold by mid-century, consistent with standard DICE projections under moderate abatement costs. The endogenous scenario

delays the 2°C crossing by a few years relative to the benchmarks, a direct consequence of its higher mitigation intensity. Figure 8 displays the corresponding emissions trajectories, which mirror the mitigation rate dynamics.

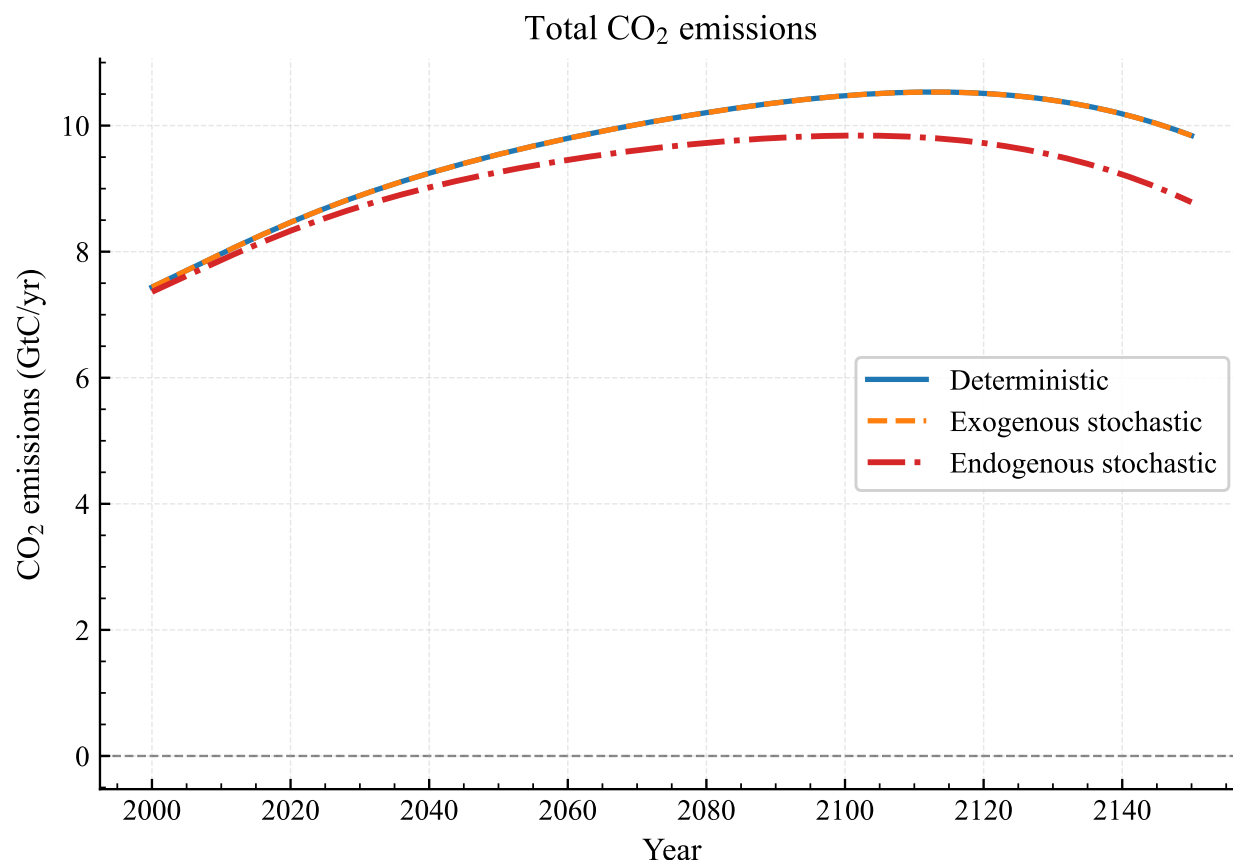


Figure 8: Total CO₂ emissions (GtC/yr) under three scenarios, 2000–2150.

Figure 9 shows the damage fraction, the share of gross output lost to climate damages, together with the 90% Monte Carlo confidence band for the endogenous scenario.

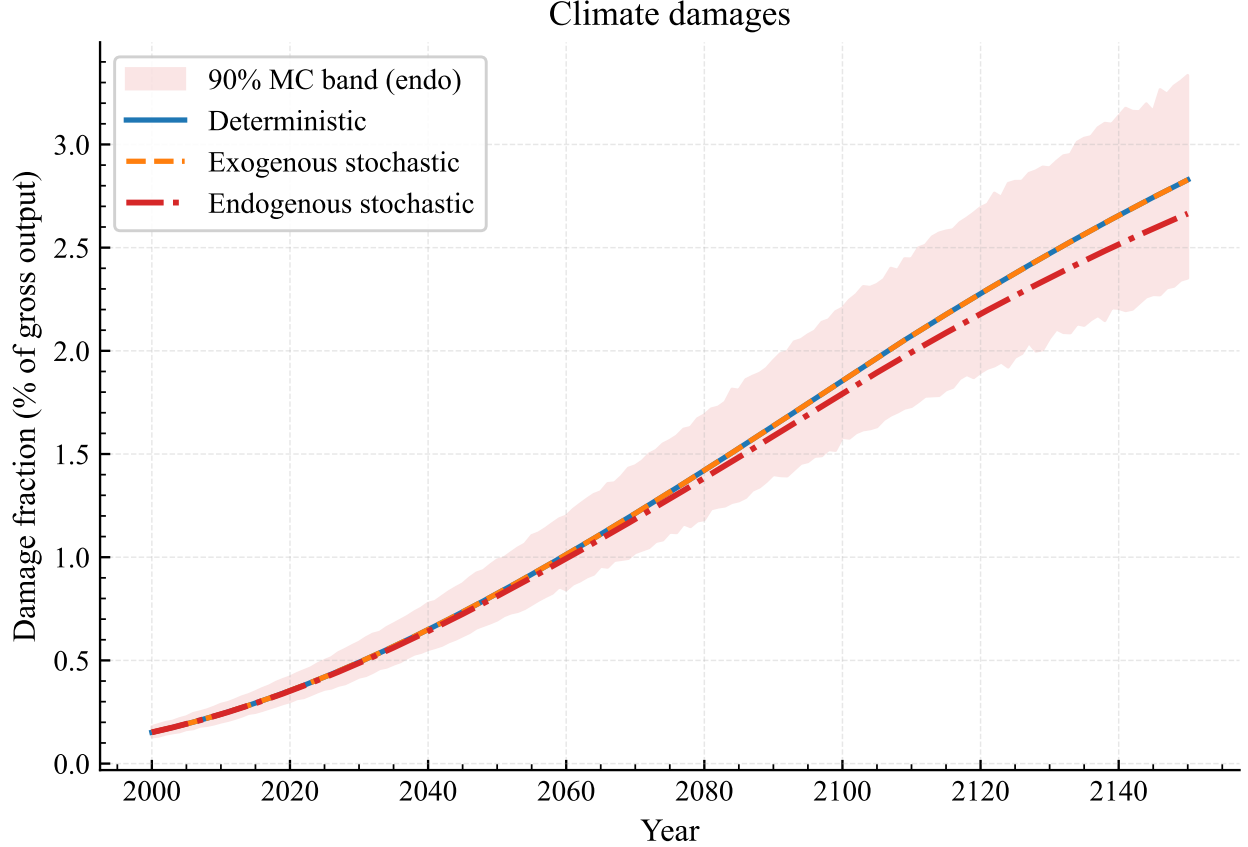


Figure 9: Damage fraction (% of gross output) under three scenarios, 2000–2150. The shaded band is an approximate 90% confidence interval from Monte Carlo simulation of the endogenous stochastic scenario ($\bar{x} \pm 1.645\hat{\sigma}$ across paths).

The Monte Carlo confidence band widening over time reflects the variance amplification channel: as CO_2 accumulates and $\sigma_x(s_t)$ grows, realized damage outcomes become increasingly dispersed. This uncertainty is precisely what the planner internalizes through the stochastic externality and what the variance channel of the optimal tax is designed to correct. Although the mean damage fractions are similar across scenarios, the endogenous scenario achieves slightly lower mean damages over the longer horizon as a result of its higher mitigation intensity.

A striking feature of the results is that per-capita consumption is virtually identical across all three scenarios at every point in time (Figure 10): all three yield approximately \$8.35k per person in 2025, \$11.46k in 2050, \$15.51k in 2075, and \$20.78k in 2100. This near-perfect equality reflects an economic balance: the higher mitigation costs borne by the endogenous scenario are approximately offset by the lower expected climate damages it achieves, leaving the average consumption path unchanged. The welfare gains from internalizing the stochastic externality therefore do not manifest as higher mean consumption but as a reduction in consumption risk, a less volatile distribution of economic outcomes with reduced exposure to severe weather events.

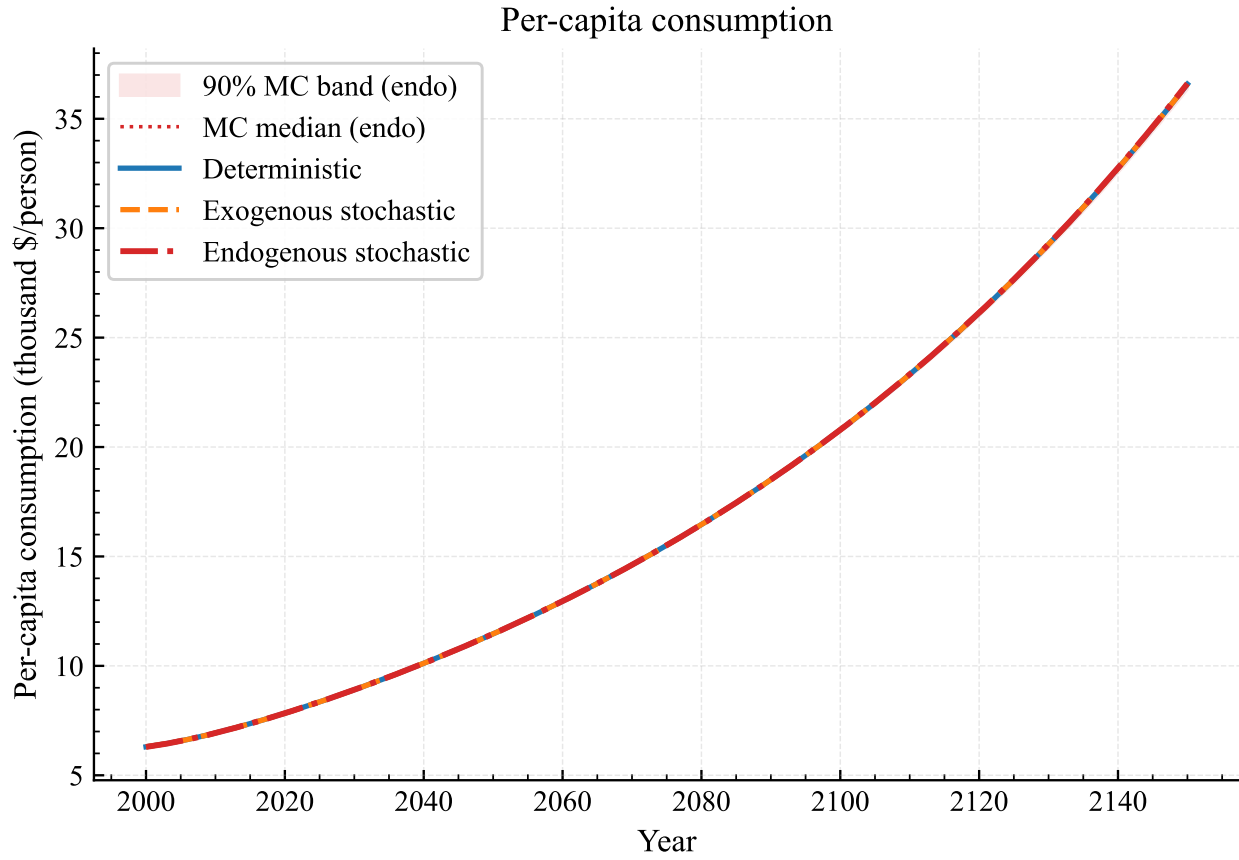


Figure 10: Per-capita consumption (k\$/person) under three scenarios, 2000–2150. The shaded band is the 90% Monte Carlo confidence interval (p5–p95) for the endogenous stochastic scenario; the dotted line is the Monte Carlo median.

6 Conclusion

This paper develops a framework for analyzing the optimal correction of stochastic externalities: the channel through which pollution shifts the entire distribution of future climate shocks rather than merely their expected value. The "distributional shift" view of climate risk that higher CO₂ concentrations make extreme weather events not only more frequent but also more severe and more unpredictable is well-documented in the Earth sciences and is beginning to be empirically quantified in economics, yet its implications for the social cost of carbon had not been formally characterized. This paper fills that gap.

Working first in a tractable three-period neoclassical model, I derive a modified optimal carbon tax that decomposes into two terms: the standard Pigouvian term and a novel stochastic externality term. The latter can be expressed as the covariance between the ratio of utilities and the score of the damage shock distribution, or equivalently as the discounted expectation of marginal damage shocks weighted by the change in the shock distribution's quantile function induced by the pollution-driven distributional shift. The framework nests the standard Pigouvian tax as a special case when the damage distribution is exogenous. Under normal shocks with pollution-dependent mean and variance, the stochastic externality decomposes into three economically interpretable channels: a mean-shift channel (pollution raises expected damages), a risk-aversion channel (pollution amplifies consumption volatility, which is costly under concave utility), and a damage-curvature channel (the interaction between increased variance and the convexity of the damage function). Under Pareto shocks, the correction collapses to a single tail-fattening channel whose magnitude can grow without bound as the tail becomes heavier, reinforcing Weitzman's Dismal Theorem: even if the standard Pigouvian term remains finite, the stochastic externality term may not. I then generalize these results to an infinite-horizon recursive model.

To illustrate the quantitative importance of the stochastic externality, I implement these results in DICE-SE, a modified version of the DICE-CJL benchmark of Cai and Lontzek (2019), in which damage shocks are drawn from a normal distribution whose mean and variance depend on atmospheric CO₂ concentration. Comparing three scenarios (deterministic, exogenous stochastic, and endogenous stochastic) reveals a clear pattern. The deterministic and exogenous scenarios are essentially equivalent: standard stochastic climate-economy models with fixed damage distributions yield a carbon price that is almost identical to the deterministic benchmark, confirming that introducing randomness without endogenizing the damage distribution has negligible policy consequences. The endogenous scenario tells a sharply different story. It prescribes a carbon price 12.3% higher than the exogenous benchmark at the initial period, with the premium growing to 14–17% throughout the 21st century as CO₂ accumulation continuously amplifies the state-dependence of the damage distribution. The optimal mitigation rate is 1.6 to 3.7 percentage points higher than the benchmark over 2000–2100, translating into approximately 0.04°C less atmospheric warming by 2100. Strikingly, per-capita consumption is virtually identical across all three scenarios: the welfare gains from internalizing the stochastic externality do not manifest as higher mean consumption but as reduced consumption risk, meaning a less volatile distribution of economic outcomes with lower exposure to severe weather events.

These findings carry direct implications for climate policy. Standard integrated assessment

models, which treat the damage shock distribution as exogenous, systematically underestimate the social cost of carbon. Under a conservative characterization using normal shocks, the carbon price should already be approximately 12% above its standard value at the outset, with the gap widening over time. Frameworks that collapse climate damages to a single expected value cannot capture this channel because the stochastic externality depends on how pollution alters the entire structure of the shock distribution.

This work requires many improvements. Currently, the DICE-SE simulation is merely illustrative, using a normal distribution with arbitrary parameters rather than a carefully-chosen realistic shock distribution (e.g. GEV, etc.), as supported by the existing empirical literature. Future work will need to properly calibrate the DICE-SE model to provide a more accurate estimate of the stochastic externality premium.

There are also a number of important extensions for future work, such as extending the framework to heterogeneous agents to allow the study of the distributional consequences of taking into account stochastic externalities. Moreover, it could be very interesting to investigate the interaction between mitigation and adaptation in this framework, as adaptation may reduce the need to internalize the stochastic externality through mitigation. Likewise, internalizing stochastic externalities may reduce the need for adaptation, acting as global insurance. Following Schumacher (2019), one might even argue that mitigating stochastic externalities is superior to adapting to them, since the former is essentially a public good while the latter is a private good, only a few being able to benefit from adaptation. In other words, a stochastic externality framework combined with an insurance component may also help us evaluate the relative merits of *ex post* mitigation ("prevention is better than cure") and of *ex ante* adaptation in the context of climate change.

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A Proof of Proposition 2.2.1

First, we solve the household's problem, which has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(r_0 k_0 - k_1 + (1 - \delta)k_0) + \beta \int_{x_1} u(r_1(x_1)k_1 - k_2(x_1) + (1 - \delta)k_1) f(x_1; \theta_1) dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(r_2(x_2)k_2(x_1) + (1 - \delta)k_2(x_1)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2\end{aligned}$$

We assume the household is atomistic and therefore does not take into account the effect of their actions on the environment. Hence, we treat the pdf as exogenous in this problem. This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + r_1) f(x_1; \theta_1) dx_1 \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + r_2) f(x_2; \theta_2) dx_2\end{aligned}$$

By the firm's optimality condition, we have $r_t = (1 - D(S_t, x_t))Y'(k_t)$. Hence,

$$\begin{aligned}u'(c_0) &= \beta \mathbb{E}_{x_1} [u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))] \\ u'(c_1) &= \beta \mathbb{E}_{x_2} [u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))]\end{aligned}$$

This is the standard Euler condition. The planner's problem yields the following FOCs.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1)) f(x_1; \theta) dx_1 + \mu_2 Y'(k_1) \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2)) f(x_2; \theta) dx_2 \\ \frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1) D_S(S_1) Y(k_1) f(x_1; \theta) dx_1 \\ & + \beta \int_{x_1} u(c_1) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_2; \theta(S_2)) dx_1 dx_2 + (1 - \phi) \mu_2 = \mu_1 \\ \frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2) D_S(S_2) Y(k_2) f(x_1; \theta) f(x_2; \theta) dx_1 dx_2 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_2; \theta)}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1)) dx_1 dx_2 = \mu_2\end{aligned}$$

So that the terms in the brackets of the first-order condition with respect to c_0 are expressed as an expectation, we would like to rewrite the equation so that the second term is multiplied by the original pdf. To do so, we can use the property that $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$.

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))f(x_1; \theta)dx_1 + \mu_2 Y'(k_1) \\
\frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))f(x_2; \theta)dx_2 \\
\frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1)D_S(S_1)Y(k_1)f(x_1; \theta)dx_1 \\
& + \beta \int_{x_1} u(c_1) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))dx_1 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 + (1 - \phi)\mu_2 = \mu_1 \\
\frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 = \mu_2
\end{aligned}$$

The last FOC can be written more compactly as follows.

$$\begin{aligned}
\mu_2 &= -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta)f(x_2; \theta)dx_1 dx_2
\end{aligned}$$

From here, we can get the following modified Euler condition between c_0 and c_1 .

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u(c_2)}{u_{c,1}} \frac{\partial \ln(f(\cdot))}{\partial S_2}}_{\text{"Stochastic" ext.}} \right] Y_{k,1} \right) \right] \quad (4)$$

If we compare this to the household's Euler condition, we see that there is a wedge on the right-hand side. In order to decentralize the planner's solution, we need to set a carbon tax

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2}}_{\text{"Stochastic" externality}} \right]$$

We can also write this in a covariance formulation⁵.

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

We can also write the Euler condition in cdf terms rather than pdf terms by rewriting the first-order condition with respect to S_2 with integration by parts⁶.

$$\begin{aligned} \int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 &= u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2 \end{aligned}$$

By the Leibniz integral rule, the derivative of the cdf with respect to its parameter is zero at the boundaries. Therefore,

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

From the resource constraint, $\frac{\partial c_2}{\partial x_2} = -D_x(S_2, x_2)Y(k_2)$.

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = \int_{\underline{x}}^{\bar{x}} u'(c_2) D_x(S_2, x) Y(k_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

⁵ $\frac{\partial \ln(f(\cdot))}{\partial \theta}$ is the score and $\mathbb{E}[S_\theta] = 0$. Hence, the second term in $\text{Cov}(x, y) = \mathbb{E}[xy] - \mathbb{E}[x] \mathbb{E}[y]$ cancels out.

⁶ Here, $u = u(c_2) \frac{\partial \theta(S_2)}{\partial S_2}$ and $dv = \frac{\partial f(\cdot)}{\partial \theta} dx$. Moreover, note that $\frac{\partial f(\cdot)}{\partial \theta} = \frac{\partial^2 F(\cdot)}{\partial x \partial \theta}$.

If we multiply and divide by $f(x_2; \theta(S_2)) \equiv \frac{\partial F(x_2; \theta(S_2))}{\partial x_2}$, we can rewrite equation (4) as:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

There are three parts to this new term: the familiar stochastic discount factor, the marginal damages of stochastic shocks on output, and the change of the cdf over the pdf. This last part is hard to interpret, so we might want to rewrite it. The quantile function is defined as:

$$Q(p; \theta(S_2)) = F^{-1}(p; \theta(S_2))$$

From here, we can also write a cdf as follows.

$$F(Q(p; \theta(S_2)); \theta(S_2)) = p$$

If we differentiate both sides by S_2 and rearrange terms,

$$\frac{\frac{\partial F(Q(p; \theta(S_2)); \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(Q(p; \theta(S_2)); \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

If we set $x = Q(p; \theta(S_2))$ in the cdf,

$$\frac{\frac{\partial F(x; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(x; \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

Hence, a different way to write equation (4) is:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

In this case, the implied corrective tax for the CDF and quantile function formulation is:

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right]$$

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

B Proof of Proposition 2.2.2

The normal PDF is $f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. In that case,

$$\frac{\partial \ln(f(\cdot))}{\partial S_2} = \left[\frac{\mu'(S_2)}{\sigma(S_2)} z + (z^2 - 1) \frac{\sigma'(S_2)}{\sigma(S_2)} \right]$$

Here, $z \sim N(0, 1)$. Hence, we can rewrite the "risk endogeneity" term as follows.

$$\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\mu'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] + \frac{\sigma'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right]$$

Because labor is assumed to be exogenous and the shock is at the last period,

$$c_2(x_2) = (1 - D(S_2, x_2))Y(k_2) + (1 - \delta)k_2 \implies \begin{cases} \frac{\partial c_2}{\partial x_2} = -\frac{\partial D(S_2, x_2)}{\partial x_2} Y(k_2) \leq 0 \\ \frac{\partial c_2}{\partial x_2^2} = -\frac{\partial^2 D(S_2, x_2)}{\partial x_2^2} Y(k_2) \leq 0 \end{cases}$$

By Stein's lemma and assuming $u_c \geq 0$ as well as $u_{cc} \leq 0$,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] &= \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial c_2}{\partial x_2} \right] \leq 0 \\ \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right] &= \sigma(S_2)^2 \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial c_2}{\partial x_2} \right)^2 + \frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 c_2}{\partial x_2^2} \right] \end{aligned}$$

Although $\frac{\partial^2 c_1}{\partial x^2}$ could be positive or negative depending on the convexity of damages, we are squaring it and therefore making the entire term positive. Since we assumed $u_{cc} \leq 0$, the first term in the second equation is negative. However, the second term's sign depends on $\frac{\partial^2 c_2}{\partial x_2^2} = -D_{xx}Y(k_2)$, the curvature of damages in x . If we rewrite the "risk endogeneity" term,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= -\mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad + \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad - \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ for $x \geq x_m$. In that case,

$$\frac{\partial \ln(f(x; \alpha(S_2)))}{\partial S_2} = \frac{\alpha'(S_2)}{\alpha(S_2)} \left[1 - \alpha(S_2) \ln \left(\frac{x}{x_m} \right) \right]$$

Let us define $y \equiv \ln \left(\frac{x}{x_m} \right)$. It is known that $y \sim \exp(\alpha)$, so we have $y \sim \exp(\alpha(S_2))$. Hence,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} - \alpha(S_2) \frac{u(c_2)}{u'(c_1)} y \right] \\ &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \right] - \alpha'(S_2) \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} y \right] \end{aligned}$$

C Proof of Proposition 3.0.1

The problem (3) has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' - \lambda [c + k' - (1 - D(S, x))Y(k) - (1 - \delta)k] \\ & - \mu [S' - (1 - \phi)S - Y(k)]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c} : & u'(c) = \lambda \\ \frac{\partial \mathcal{L}}{\partial k'} : & \beta \int \frac{\partial V}{\partial k'} f(x'; \theta(S')) dx' = \lambda \\ \frac{\partial \mathcal{L}}{\partial S'} : & \beta \int \frac{\partial V}{\partial S'} f(x'; \theta(S')) dx' + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx' = \mu\end{aligned}$$

By the envelope condition,

$$\frac{\partial V}{\partial S'} = -\lambda' D_S(S', x') Y(k') + (1 - \phi)\mu'$$

If we substitute λ' ,

$$\frac{\partial V}{\partial S'} = -u'(c') D_S(S', x') Y(k') + (1 - \phi)\mu'$$

Substituting into the FOC with respect to S' ,

$$\begin{aligned}\mu = & -\beta \int [u'(c') D_S(S', x') Y(k') - (1 - \phi)\mu'] f(x'; \theta(S')) dx' \\ & + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx'\end{aligned}$$

Using $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$, we can put everything under the same expectation.

$$\mu = -\beta \int \left[u'(c') D_S(S', x') Y(k') - V(k', S', x') \frac{\partial \ln(f(x'; \theta(S')))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} - (1 - \phi) \mu' \right] f(x'; \theta(S')) dx'$$

By the envelope condition,

$$\begin{aligned} \frac{\partial V}{\partial k'} &= \lambda' (1 - \delta + (1 - D(S', x')) Y'(k')) + \mu' Y'(k') \\ &= \lambda' \left(1 - \delta + \left(1 - D(S', x') + \frac{\mu'}{\lambda'} \right) Y'(k') \right) \end{aligned}$$

Hence, after combining the first two FOCs,

$$\begin{aligned} u'(c) &= \beta \int u'(c') \left(1 - \delta + \left(1 - D(S', x') - \beta \int \left[\frac{u'(c'')}{u'(c')} D_S(S'', x'') Y(k'') \right. \right. \right. \\ &\quad \left. \left. \left. - \frac{V(k'', S'', x'')}{u'(c')} \frac{\partial \ln(f(x''; \theta(S'')))}{\partial S''} \right. \right. \right. \\ &\quad \left. \left. \left. - (1 - \phi) \frac{\mu''}{u'(c')} \right] f(x''; \theta(S'')) dx'' \right) Y'(k') \right) f(x'; \theta(S')) dx' \end{aligned}$$

By induction, we can put every $u'(\cdot)$ term and V term together in a weighted infinite sum. Meanwhile, the $(1 - \phi)\mu$ term vanishes by the transversality condition for the condition stock since $\beta(1 - \phi) < 1$ and μ is bounded.

$$\begin{aligned} u'(c_t) &= \beta \int u'(c_{t+1}) \left(1 - \delta + \left(1 - D_{t+1} - \beta \int \left[\sum_{j=0}^{\infty} ((1 - \phi)\beta)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_S(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \right. \right. \right. \\ &\quad \left. \left. \left. - ((1 - \phi)\beta)^j \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f(x_{t+j+2}; \theta(S_{t+j+2})))}{\partial S_{t+j+2}} \right] f(\cdot) \right) Y'(k_{t+1}) \right) f(x_{t+1}; \theta(S_{t+1})) dx_{t+1} \end{aligned}$$

Hence, the corrective tax for period t is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right] \right]$$

In covariance form,

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] - \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)$$

We can also write the stochastic term in CDF or quantile form. Consider the stochastic part of the tax for a given j :

$$\int V(k_{t+j+2}, S_{t+j+2}, x_{t+j+2}) \frac{\partial \theta(S_{t+j+2})}{\partial S_{t+j+2}} \frac{\partial f(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial \theta} dx_{t+j+2}$$

Integration by parts with $u = V_{t+j+2} \frac{\partial \theta}{\partial S}$ and $dv = \frac{\partial f}{\partial \theta} dx$ yields:

$$\begin{aligned} &= V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} \frac{\partial V_{t+j+2}}{\partial x_{t+j+2}} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2} \end{aligned}$$

Here, the first term vanishes by the boundary condition. By the envelope theorem,

$$\frac{\partial V(k, S, x)}{\partial x} = -u'(c) D_x(S, x) Y(k)$$

Therefore,

$$\int V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial f}{\partial \theta} dx_{t+j+2} = \int u'(c_{t+j+2}) D_x(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2}$$

Multiplying and dividing by f , the CDF form of the tax is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial S_{t+j+2}}}{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial x_{t+j+2}}} \right] \right]$$

Applying the quantile identity,

$$\frac{\frac{\partial F(x; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}}{f(x; \theta(S))} = - \frac{\partial Q(p; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}$$

Hence, the quantile form of the tax is:

$$\begin{aligned} \tau_t = & \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] \\ & + \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right] \end{aligned}$$